

BAULKHAM HILLS HIGH SCHOOL



MATHEMATICS EXTENSION 1 ASSESSMENT

December 2011

*Time allowed: 50 minutes
plus 5 minutes reading time*

STUDENT NUMBER : _____

TEACHER'S NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
TOTAL	
PERCENTAGE	



Extension-1 Mathematics

December 2011

Time: 50 minutes + 5 minutes reading time

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- At the end of the exam, staple your answers in order behind the cover sheet provided, and your questions on the back
- Approved Maths aids and calculators may be used

1.	Determine whether each statement is true or false for $n = 1$. a) $5 + 10 + x + \dots + 5(2^{n-1}) = 5(2^n - 1)$ b) $\sum_{n=1}^r 64(-2)^{1-n} = 32(1 - 2^{-r})$	2
2.	Determine which of the following expressions are polynomials a) $x^4 - 2x + 9$ b) $4x^3 - 5x + \log x$	1 1
3.	What is the degree of the polynomial $P(x) = (4x - 6)(3x^4 - 2x) + 12$	1
4.	A polynomial of degree 3 has zeros at $-1, 2$ and 3 . Its leading coefficient is 4 . What is the constant term.	2
5.	a) Without using calculus, sketch the graph $f(x) = x^3 - 5x^2 - 14x$ b) Hence (or otherwise) solve $\frac{x^2 - 4x - 14}{x} \geq 1$	2 2
6.	If α, β and γ are the roots of the polynomial $x^3 + 2x^2 - x - 5 = 0$ Find a) $\alpha + \beta + \gamma$ b) $\alpha\beta + \beta\gamma + \alpha\gamma$ c) $\alpha\beta\gamma$ d) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ e) $(\alpha + 1)(\beta + 1)(\gamma + 1)$	1 1 1 2 2
7.	Using mathematical induction prove that for integers $n \geq 1$ $\sum_{r=1}^n r \times 2^{r-1} = (n-1)2^n + 1$	3
8.	Find the roots of the following polynomial $4x^3 - 4x^2 - 29x + 15 = 0$ given that one root is the difference between the other two roots.	3
9.	Use the principle of mathematical induction to prove that $5^n + 12n - 1$ is a multiple of 16 for all positive integers n	3
10.	The polynomial $P(x) = 4x^3 + ax^2 + 3x + b$ is divisible by $(x + 2)$ and $(x - 1)$ Find a and b .	3

11.	Given that $(x - 5)$ is a factor of $P(x) = -x^3 - x^2 + 21x + 45$	3
	a) Express $P(x)$ in factored form b) Without the aid of calculus, sketch $P(x)$ showing main features.	2
12.	The polynomial $f(x)$ • Has degree 37 • A remainder of 1 when divided by $(x - 1)$ • A remainder of 3 when divided by $(x - 3)$ • A remainder of 21 when divided by $(x - 5)$ Find the remainder when $f(x)$ is divided by $(x - 1)(x - 3)(x - 5)$	3
~ END OF EXAM ~		

1. a) T ✓

b) F ✓

2 a) Yes ✓

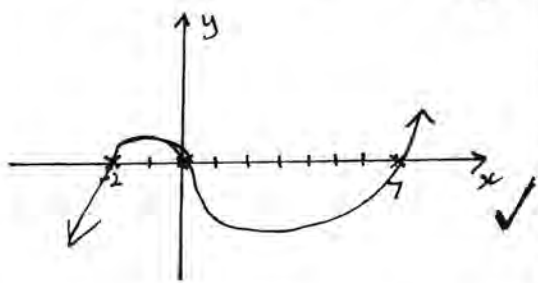
b) NO ✓

3. 5 ✓

4. 24 ✓✓

$4(x+1)(x-2)(x-3)$

5 a) $f(x) = x(x-7)(x+2)$ ✓



b) $\frac{x^2-4x-14}{x} - 1 \geq 0$

$x^2 \left(\frac{x^2-5x-14}{x} \right) \geq 0 \times x^2$

$\therefore x(x^2-5x-14) \geq 0$ ✓✓

$\therefore -2 \leq x < 0, x \geq 7$

b. $x^3 + 2x^2 - x - 5 = 0$

a) sum: $-\frac{b}{a} = -2$ ✓

b) sum $\times 2$: $\frac{c}{a} = -1$ ✓

c) product: $-\frac{d}{a} = 5$ ✓

d) $\frac{\beta\gamma + \alpha\beta + \alpha\gamma}{2\beta\gamma} = -\frac{1}{5}$ ✓✓

e) $\alpha\beta\gamma + \alpha\gamma + \beta\gamma + \gamma + \alpha\beta + \alpha + \beta$ ✓

$= 5 - 1 - 2 + 1$

$= 2 + 1 = 3$ ✓

⑦ $\sum_{r=1}^n r \times 2^{r-1} = (n-1)2^n + 1$

✓ structure

✓ step 3.

$S_n = 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = (n-1)2^n + 1$

Test true for $n=1$

LHS = 2^0 RHS = $0 \times 2^1 + 1$

$= 1$ $= 1$ $\therefore$ true for $n=1$.

assume true for $n=k$.

$S_k = 1 + 2 \times 2 + 3 \times 2^2 + \dots + k \times 2^{k-1} = (k-1)2^k + 1$

prove true for $n=k$.

ie $S_k + T_{k+1} = S_{k+1}$

prove $(k-1)2^k + 1 + (k+1)2^k = (k)2^{k+1} + 1$

LHS = $k \cdot 2^k - \cancel{2^k} + 1 + k \cdot 2^k + \cancel{2^k}$

$= 2 \cdot k \cdot 2^k + 1$

$= (k)2^{k+1} + 1$

$=$ RHS.

$\therefore$ if true for $n=k$ now true for $n=k+1$

Since true for $n=1$ then true for $n=1+1=2$ and $n=2+1=3$ and so on by m.I for all integers $n \geq 1$

⑧ $4x^3 - 4x^2 - 29x + 15 = 0$

let $\gamma = \alpha - \beta$

sum: $\alpha + \beta + \alpha - \beta = -\frac{b}{a}$

product: $\alpha\beta(\alpha - \beta) = -\frac{15}{4}$

$2\alpha = 1$ ✓
 $\alpha = \frac{1}{2}$

$\frac{1}{2}\beta(\frac{1}{2} - \beta) = -\frac{15}{4}$

$2\beta^2 - \beta - 15 = 0$ ✓

$(2\beta + 5)(\beta - 3) = 0$

$\beta = -\frac{5}{2}$ $\beta = 3$

$\therefore$ check.

$\gamma = \frac{1}{2} - -\frac{5}{2}$

$= 3$

$\gamma = \frac{1}{2} - 3$

$= -\frac{5}{2}$

$\therefore$ zeros or roots are $\frac{1}{2}, -\frac{5}{2}, 3$. ✓

9. $5^n + 12n - 1$ is a multiple of 16 ✓✓
 test true for $n=1$
 $5^1 + 12 - 1 = 16$ which is divisible by 16.

assume true for $n=k$

$$\frac{5^k + 12k - 1}{16} = m \quad (\text{where } m \text{ is some integer})$$

$$\therefore 5^k + 12k - 1 = 16m$$

$$\text{or } 5^k = 16m - 12k + 1 \dots \textcircled{1}$$

prove true for $n=k+1$

$$\text{prove } 5^{k+1} + 12(k+1) - 1 = 16N$$

(where N is some integer)

$$\text{now LHS} = 5^k \cdot 5 + 12k + 12 - 1$$

$$\begin{aligned} \text{using } \textcircled{1} &= (16m - 12k + 1)5 + 12k + 11 \\ &= 5 \times 16m - 60k + 5 + 12k + 11 \\ &= 5 \times 16m - 48k + 16 \\ &= 16(5m - 3k + 1) \end{aligned}$$

which is a multiple of 16 for $N = 5m - 3k + 1$

$\therefore$ if true for $n=k$ now true for $n=k+1$
 Since true for $n=1$ now true for $n=1+1=2$, $n=2+1=3$ and so on by M.I.
 for all $n \geq 1$.

10. $P(x) = 4x^3 + ax^2 + 3x + b$

$$P(-2) = 4 \times (-2)^3 + a \times 4 - 6 + b = 0$$

$$4a + b = 38 \dots \textcircled{1}$$

$$P(1) = 4 + a + 3 + b = 0$$

$$a + b = -7 \dots \textcircled{2}$$

$$\textcircled{1} - \textcircled{2} \quad 3a = 45$$

$$a = 15 \quad \checkmark$$

sub back into $\textcircled{2}$

$$15 + b = -7$$

$$b = -22$$

$$\therefore a = 15, b = -22. \quad \checkmark$$

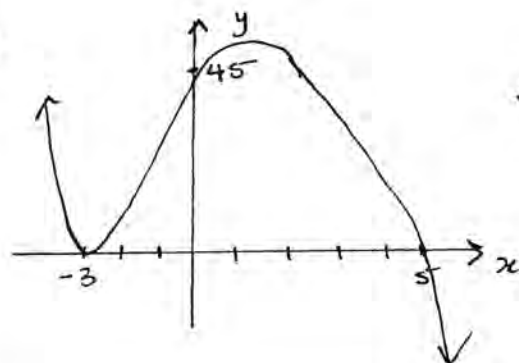
$$\begin{array}{r} -x^2 - 6x - 9 \\ x-5 \overline{) -x^3 - x^2 + 21x + 45} \\ \underline{-x^3 + 5x^2} \\ -6x^2 + 21x \\ \underline{-6x^2 + 30x} \\ -9x + 45 \\ \underline{-9x + 45} \\ 0 \end{array}$$

✓✓
working

$$\therefore P(x) = (x-5)(-x^2 - 6x - 9)$$

$$= -(x-5)(x+3)^2 \quad \checkmark$$

b)



✓✓

12. $P(x) = (x-1)(x-3)(x-5)Q(x) + R(x)$
 $R(x)$ could be $ax^2 + bx + c$ to have degree less than $A(x)$ ✓
 $\therefore R(1) = a + b + c = 1 \dots \textcircled{1}$ ✓
 $R(3) = 9a + 3b + c = 3 \dots \textcircled{2}$
 $R(5) = 25a + 5b + c = 21 \dots \textcircled{3}$

$$\textcircled{2} - \textcircled{1} \quad 8a + 2b = 2 \dots \textcircled{4}$$

$$\textcircled{3} - \textcircled{2} \quad 16a + 2b = 18 \dots \textcircled{5}$$

$$\textcircled{5} - \textcircled{4} \quad 8a = 16$$

$$a = 2$$

sub into $\textcircled{4}$

$$16 + 2b = 2$$

$$2b = -14$$

$$b = -7$$

sub into $\textcircled{1}$

$$2 - 7 + c = 1$$

$$c = 6$$

remainder

$$R(x) = 2x^2 - 7x + 6 \quad \checkmark$$

3

3